

AudioSurvey AI

AI-Powered Multilingual IVR Voice Survey Platform

Version 1.1.0 | Last Updated: 2026-03-20 | Classification: Internal

Project Documentation

Revision History

Version	Date	Author	Change Description
1.0.0	2026-02-22	Krishnanand	Initial release — core IVR engine, Twilio integration, Whisper STT, admin dashboard
1.0.1	2026-02-26	Krishnanand	Excel export support for survey responses
1.0.2	2026-03-01	Krishnanand	Excel formatting corrections, UX/UI upgrade, audio channel fix for full-call recordings
1.0.3	2026-03-03	Krishnanand	Updated survey questions, system cleanup
1.0.4	2026-03-04	Krishnanand	English translation export, "Convert to English" button added
1.0.5	2026-03-05	Krishnanand	Authentication system — login/logout, brute-force protection, session management
1.0.6	2026-03-05	Krishnanand	Full call testing, results saved and verified end-to-end
1.0.7	2026-03-08	Krishnanand	MCQO digit bug fix, hardcoded button logic corrected
1.0.8	2026-03-10	Krishnanand	DeepFilterNet background noise removal, random crash fix, log improvements
1.0.9	2026-03-15	Krishnanand	macOS DMG packaging, icon generator update
1.1.0	2026-03-17	Krishnanand	File naming convention refactored, auth info cleanup, documentation

Table of Contents

#	Section	Description
1	Executive Summary	High-level project overview and objectives
2	System Architecture	Architecture diagrams, component topology, tech stack
3	Component Design	Deep dive into each module and its responsibilities
4	Data Flow & Pipelines	End-to-end data lifecycle and ML processing pipeline
5	API Reference	Complete HTTP endpoint documentation
6	Data Models & State Schema	JSON schemas, state machine, data persistence
7	Authentication & Security	Auth flow, brute-force protection, session management
8	Survey Question Engine	Question format specification and IVR call flow
9	Configuration Reference	All configuration files and environment variables

#	Section	Description
10	Admin Dashboard	UI features, live polling, user workflows
11	Background Services	Scheduler, ML worker, and concurrency model
12	Error Handling & Resilience	Fault tolerance, retry logic, graceful degradation
13	Logging & Observability	Logging architecture, audit trails, monitoring
14	Deployment Guide	Local setup, ngrok tunneling, macOS DMG packaging
15	Directory Structure	Complete project tree with annotations
16	Dependency Matrix	Third-party libraries and their roles
17	Glossary	Domain terminology and abbreviations

1. Executive Summary

1.1 Purpose

AudioSurvey AI is an AI-powered multilingual Interactive Voice Response (IVR) survey platform designed to conduct automated voice-based research surveys over telephone calls. The system targets **African refugees who speak Kiswahili**, enabling researchers to collect structured survey responses at scale without requiring in-person enumerators.

1.2 Business Context

The platform was built for academic and public health research. It is designed as a **flexible, multi-survey system** — the survey topic, questions, and thematic sections are fully configurable via the `data/questions.txt` file. Researchers can deploy different surveys for different studies without any code changes.

1.3 Key Capabilities

Capability	Description
Automated Outbound Calling	Scheduled batch dialing of research participants via Twilio
Inbound Call Support	Participants can call in and complete the survey
Multilingual IVR	Survey prompts spoken in Kiswahili via Azure Neural TTS
Multi-Format Questions	INFO, OPEN (speech), MCQ (DTMF keypad), MCQO (MCQ + "Other" speech)
Full-Call Recording	Complete audio capture of every call (up to 30 minutes)
AI Audio Processing	Background noise removal via DeepFilterNet
Speech-to-Text	Post-call transcription using OpenAI Whisper (large-v3)
Machine Translation	Automatic Kiswahili to English translation
English Audio Generation	TTS synthesis of translated responses
Structured Data Export	Excel export of MCQ/MCQO responses (original + English)
Admin Dashboard	Real-time web UI for managing participants, scheduling, and monitoring
Conference Calling	Three-way call support for researcher-moderated interviews
macOS Distribution	Packaged as a native .app inside a DMG installer

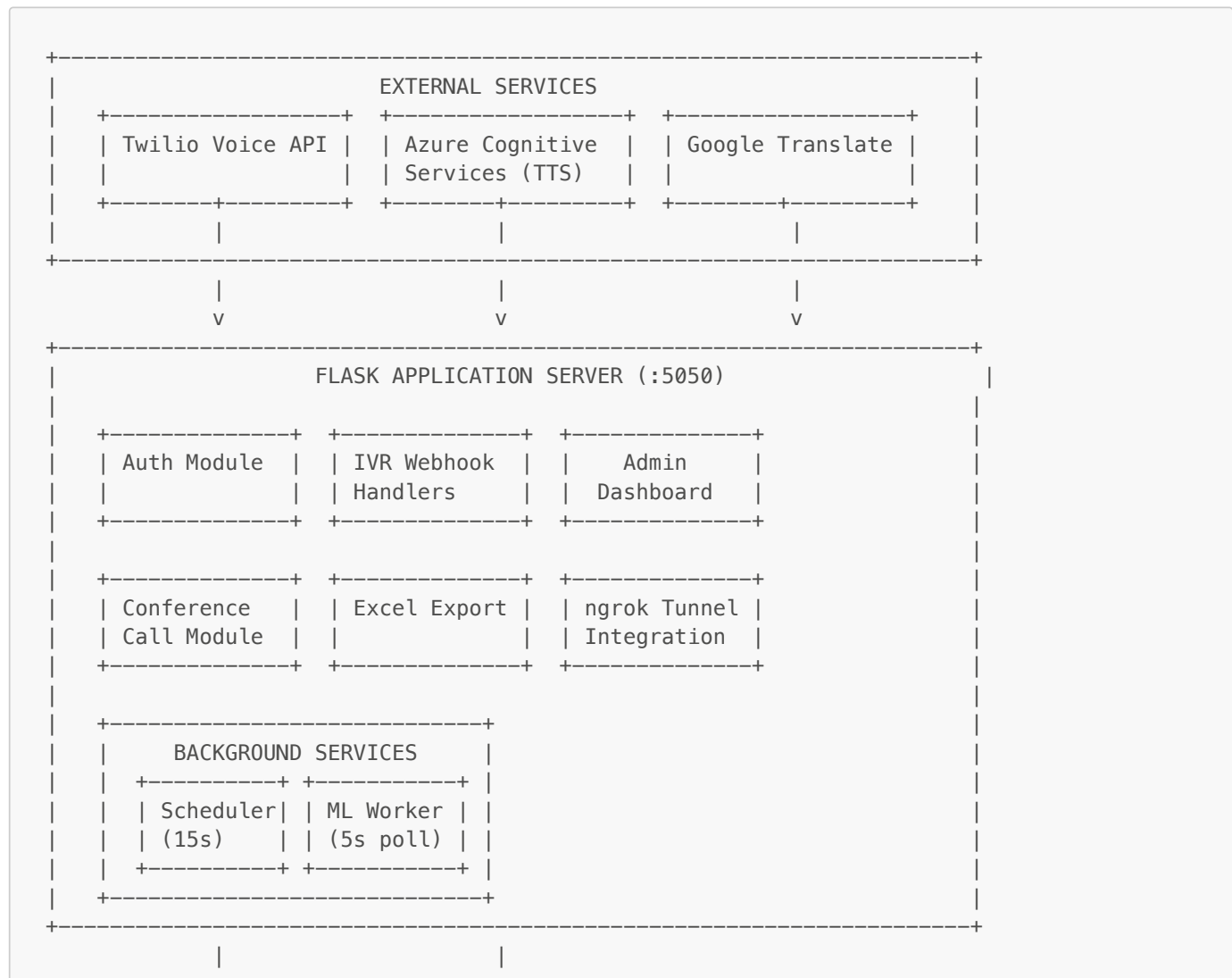
1.4 Technology Summary

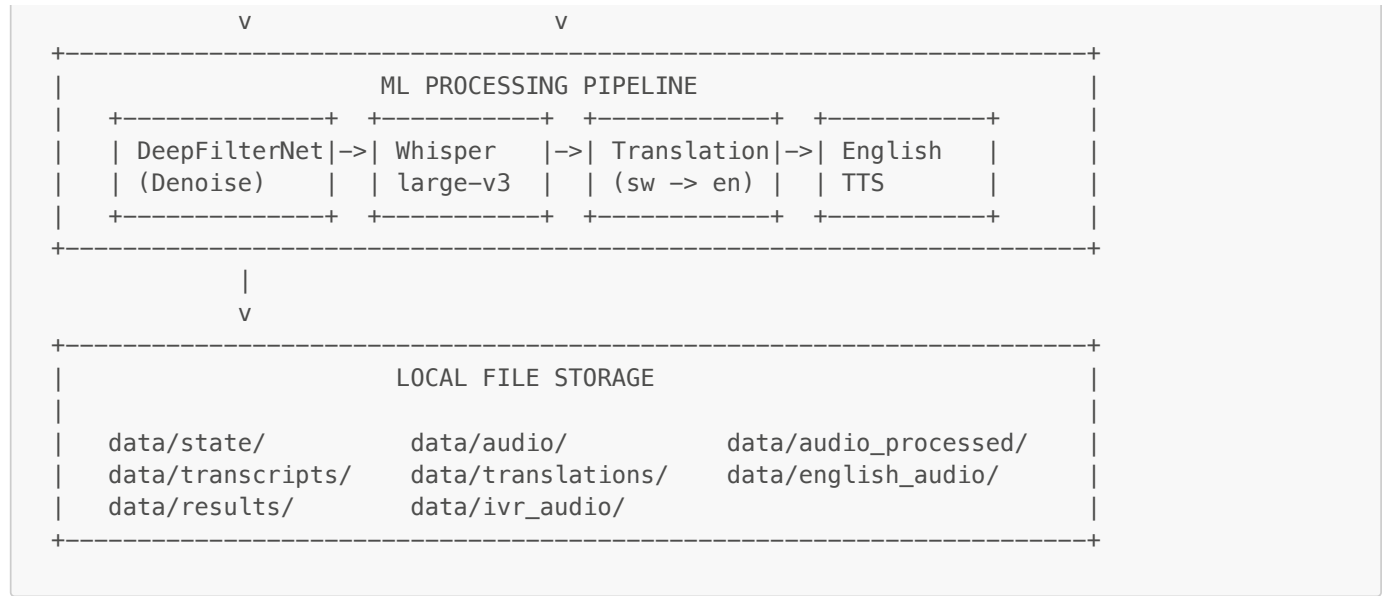
Layer	Technology
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Layer	Technology
Runtime	Python 3.x
Web Framework	Flask 3.1.2
Telephony	Twilio Voice API
Live Speech Recognition	Twilio <Gather> (Kiswahili sw-KE)
Post-Call Transcription	OpenAI Whisper large-v3
Text-to-Speech (Prompts)	Azure Cognitive Services (Neural SSML)
Text-to-Speech (Output)	Google TTS (gTTS)
Translation	Google Translate (googletrans)
Noise Reduction	DeepFilterNet (PyTorch-based)
Audio Processing	FFmpeg, pydub
Data Export	pandas + openpyxl
Tunneling	ngrok
Packaging	macOS .app bundle + DMG

2. System Architecture

2.1 High-Level Architecture

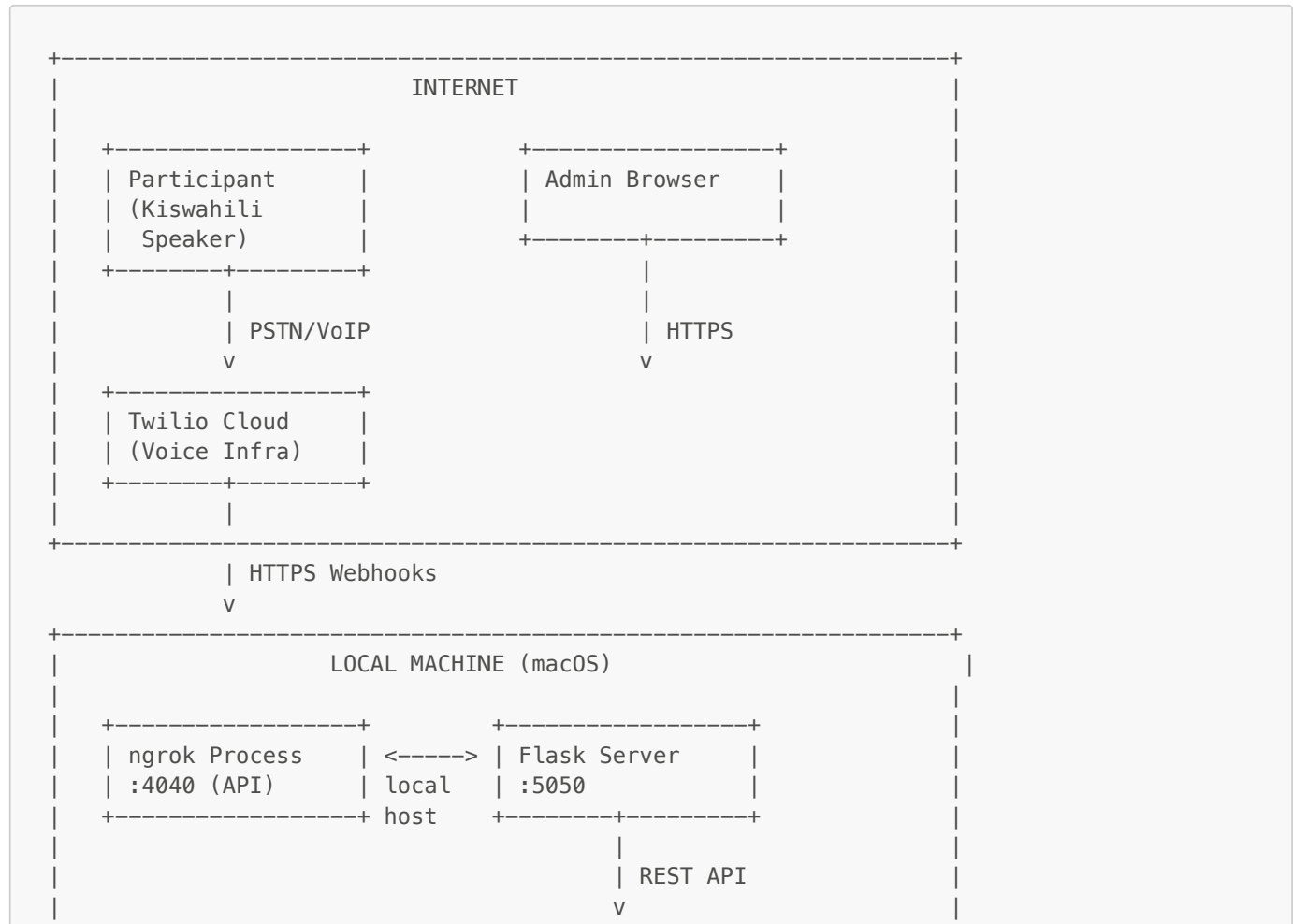




Data flow summary:

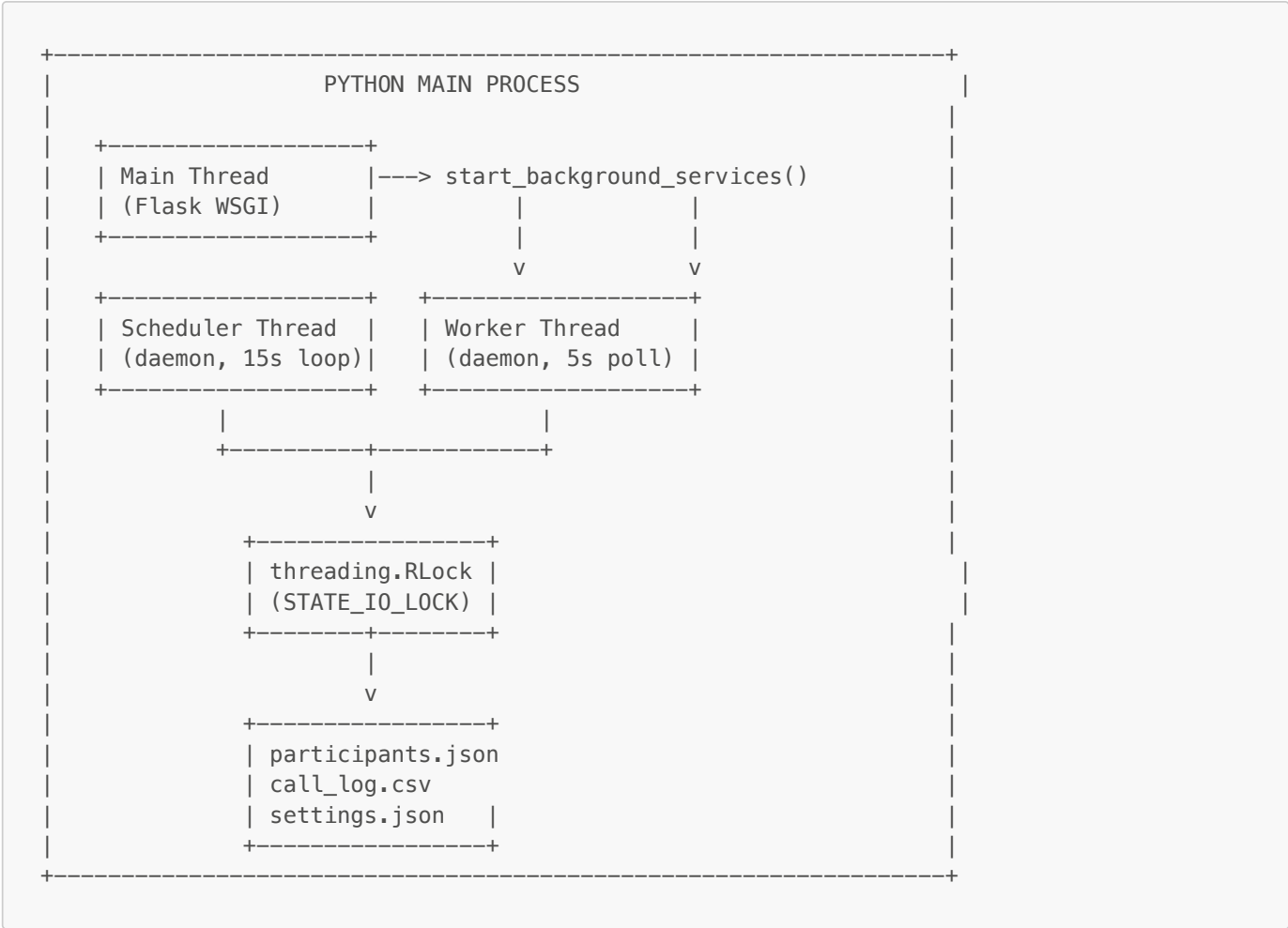
- Participant Phone <--> Twilio <--> ngrok <--> IVR Webhook Handlers
- Admin Browser --> ngrok --> Auth Module --> Dashboard
- IVR Handlers --> Azure TTS --> data/ivr_audio/
- IVR Handlers --> Recording Done --> ML Worker
- ML Worker --> DeepFilterNet --> Whisper --> Translation --> English TTS
- Translation --> Google Translate API
- Scheduler --> Twilio (dial eligible participants)
- Excel Export --> data/state/ --> data/results/

2.2 Network Topology



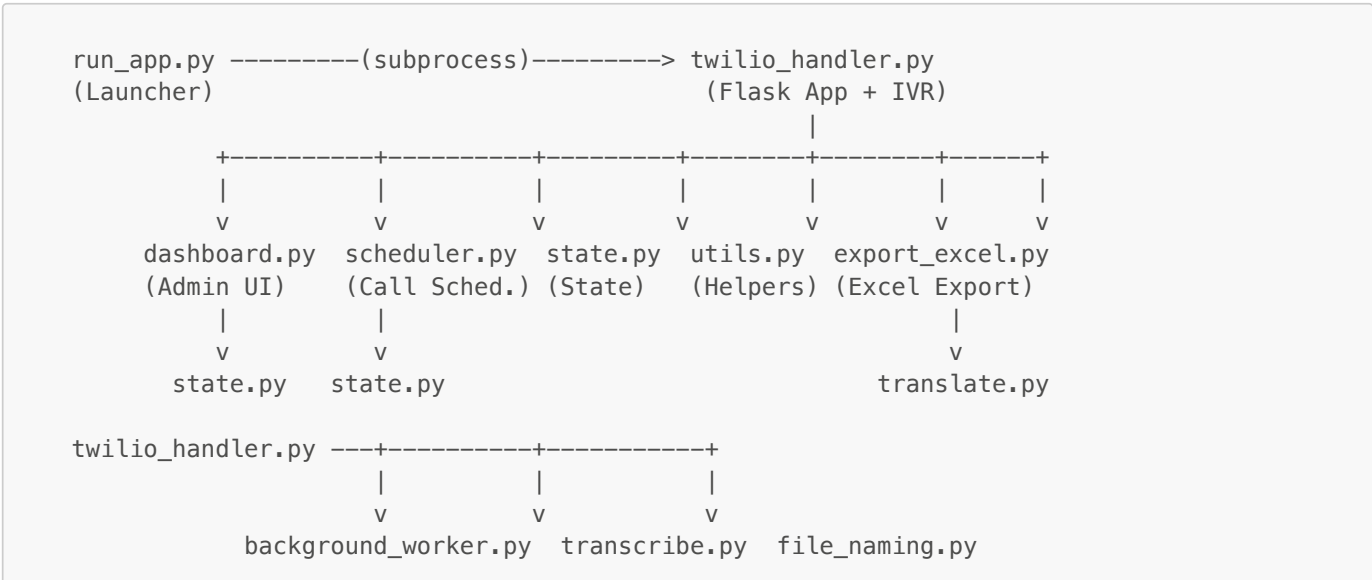


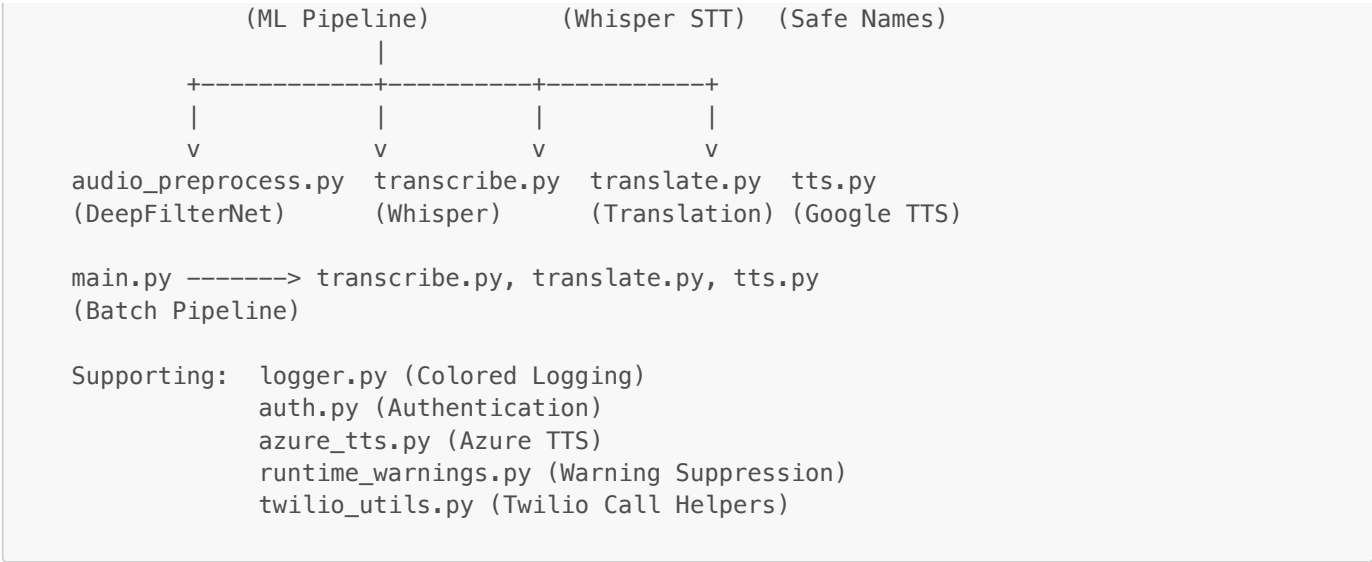
2.3 Thread Architecture



3. Component Design

3.1 Module Dependency Graph





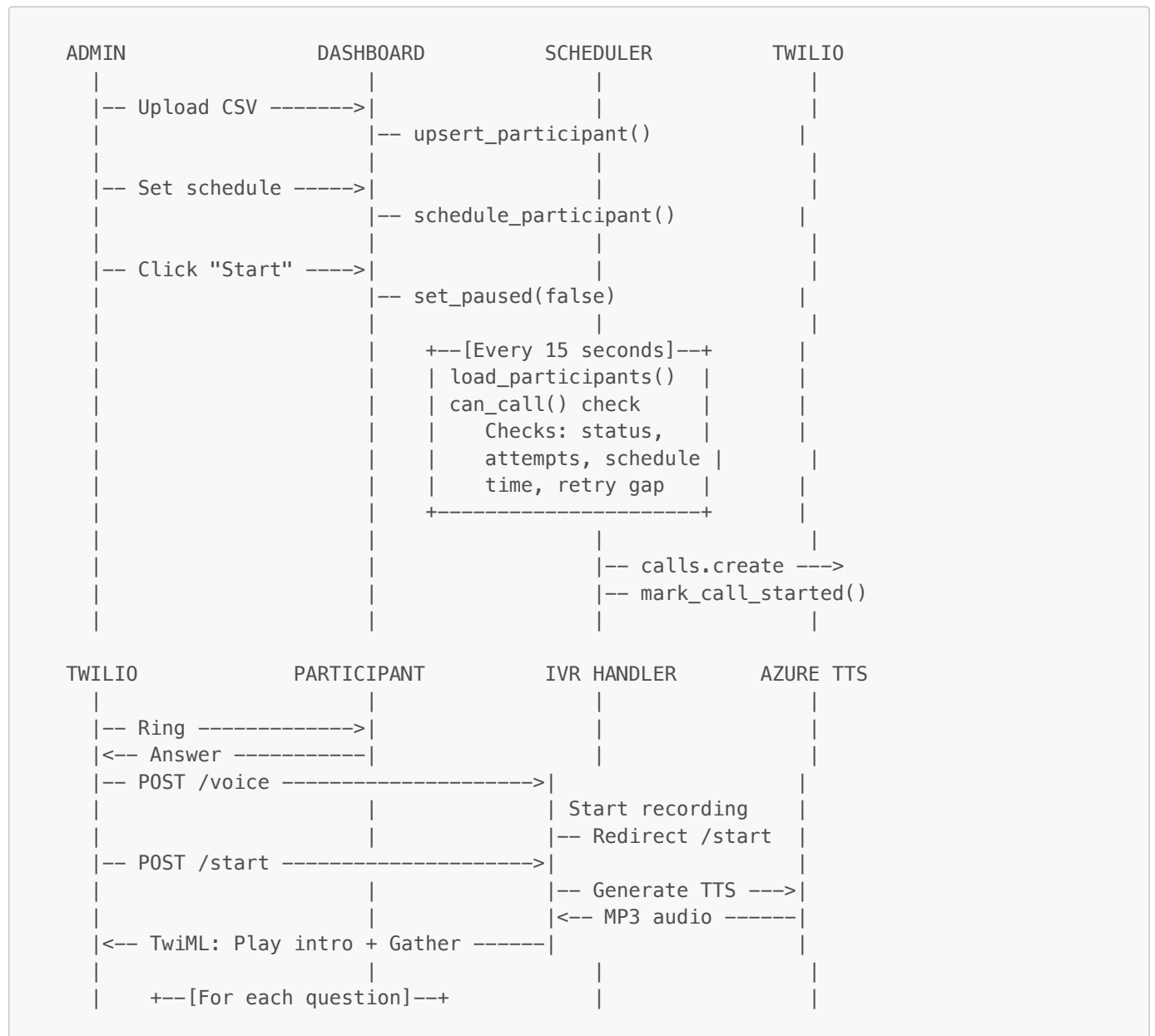
3.2 Module Descriptions

Module	Lines	Responsibility
twilio_handler.py	~1,330	Core Flask application. Defines all IVR webhook routes, Azure TTS for call prompts, auth login/logout flows, conference calling, recording download, and application bootstrap. Acts as the central orchestrator.
dashboard.py	~950	Admin web dashboard. Renders the single-page HTML UI with inline CSS/JS. Handles participant management routes (upload, schedule, pause/resume, reset, dial now). Includes live-polling JSON endpoint.
state.py	~240	Thread-safe participant state management. Handles JSON persistence with atomic writes, call eligibility logic (can_call), state transitions, retry gap enforcement, and participant schema migration.
export_excel.py	~310	Builds structured Excel exports from participant responses. Decodes DTMF digits back to option text, filters out OPEN responses, supports both original-language and English-translated exports.
background_worker.py	~150	Continuous polling worker that processes completed recordings through the 4-stage ML pipeline (denoise, transcribe, translate, TTS). Shows terminal progress bars.
scheduler.py	~115	Timed call dispatcher. Runs every 15 seconds, checks participant eligibility, and places Twilio calls for scheduled participants. Supports force-dial mode.
audio_preprocess.py	~210	Audio noise removal pipeline using DeepFilterNet. Handles FFmpeg-based channel extraction/resampling, PyTorch-based noise removal, and output resampling for Whisper.
transcribe.py	~50	Whisper large-v3 speech-to-text. Transcribes audio files with Swahili language hint, returns text and detected language. Supports both single-file and directory batch modes.
translate.py	~135	Chunked Google Translate integration. Splits long texts at sentence boundaries, retries failed chunks, marks failures with placeholder tags.
tts.py	~55	Google TTS (gTTS) wrapper for generating English MP3 audio from translated text. Skips files with translation failure markers.
auth.py	~165	Standalone authentication module (partially duplicated in twilio_handler.py). Provides brute-force protection, session management, and credential verification.

Module	Lines	Responsibility
utils.py	~25	Scheduling utility. Converts NYC local time to UTC and updates participant state.
azure_tts.py	~50	Standalone Azure TTS module with disk caching. Used for generating call prompt audio.
file_naming.py	~28	Safe filename generation. Sanitizes participant IDs and generates timestamped base names.
logger.py	~75	Colored console logging via <code>colorlog</code> . Silences noisy third-party library logs. Provides context manager for extra-quiet operations.
runtime_warnings.py	~15	Suppresses urllib3 OpenSSL compatibility warnings.
run_app.py	~130	Application launcher. Auto-starts ngrok, sets environment variables, launches Flask, opens browser.
main.py	~28	Standalone batch processor for offline audio to transcript to translation to TTS pipeline.

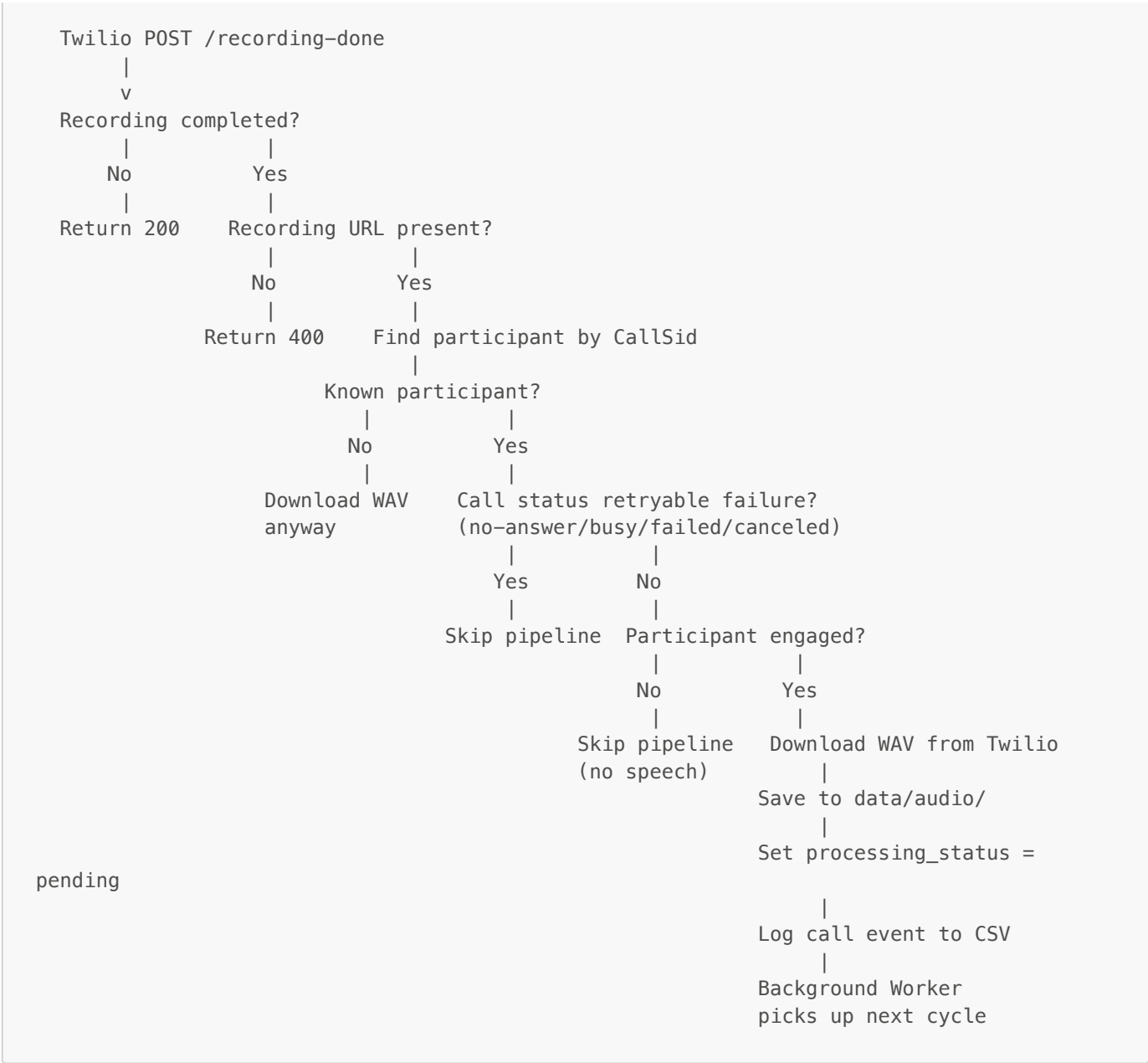
4. Data Flow & Pipelines

4.1 Outbound Call Lifecycle

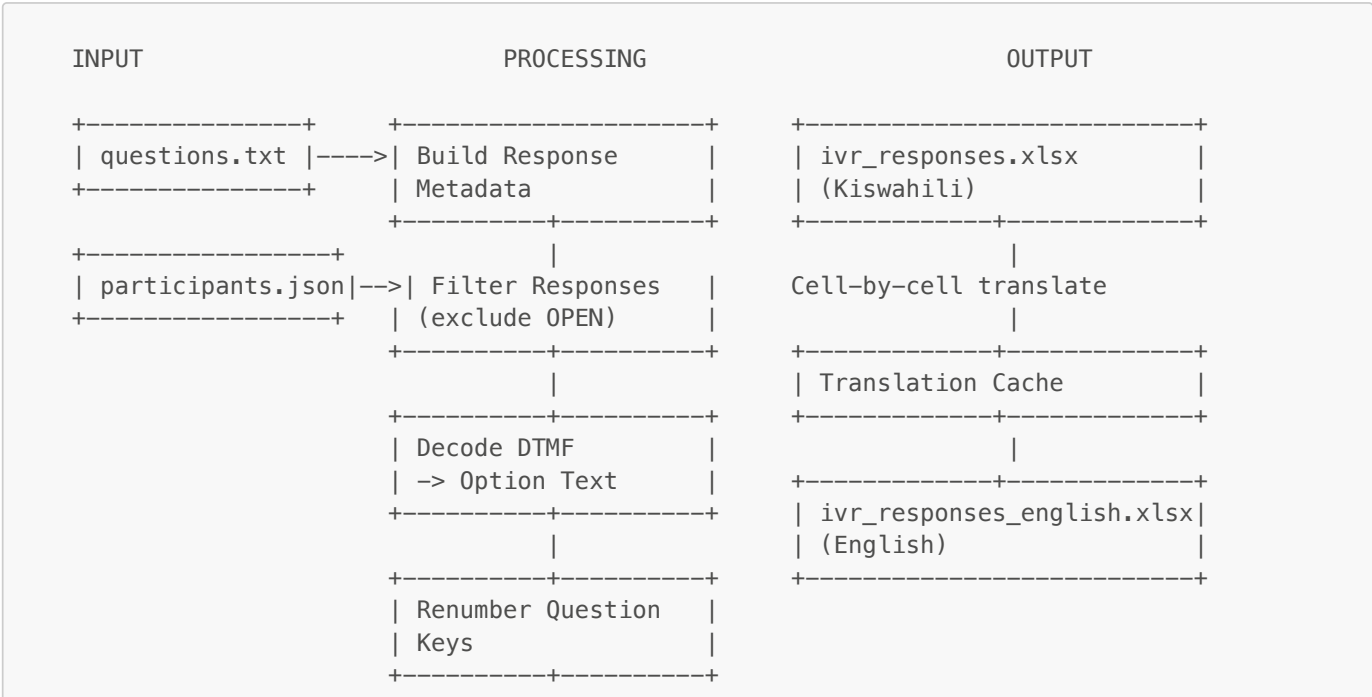


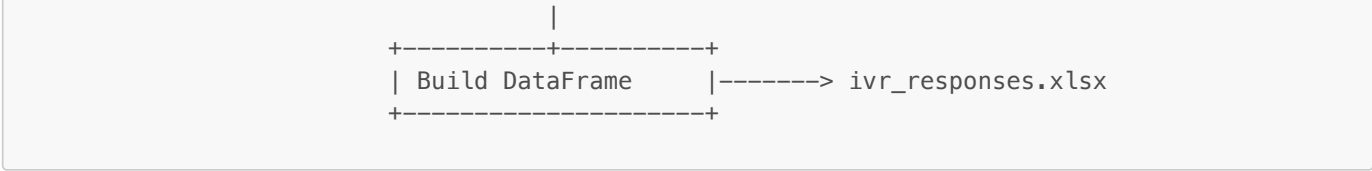
4.2 ML Processing Pipeline

4.3 Recording Callback Flow



4.4 Excel Export Data Flow





5. API Reference

5.1 Public Endpoints (No Auth Required)

These endpoints are accessible without authentication — they are called by Twilio webhooks or used for health checks.

POST /voice

Twilio Inbound Call Entrypoint

Initiates full-call recording and redirects to the survey start.

Parameter	Source	Description
CallSid	Twilio	Unique call identifier
From	Twilio	Caller phone number

Response: TwiML — starts <Recording>, redirects to /start

POST /start

Survey Introduction

Plays the first 2 intro prompts (INFO questions), then begins the first survey question.

Parameter	Source	Description
CallSid	Twilio	Call identifier

Response: TwiML — plays intros, gathers first OPEN question

POST /next?q={index}

Survey Question Router

Core question loop. Stores previous answer, advances to next question based on type.

Parameter	Source	Description
q	Query string	0-based question index
CallSid	Twilio	Call identifier
SpeechResult	Twilio	Transcribed speech (OPEN questions)

Response: TwiML — varies by question type (Gather speech / Gather DTMF / Play + Redirect)

POST /mcq-handler?q={index}

MCQ/MCQO DTMF Handler

Processes keypad digit for multiple choice questions. If MCQO and "Other" selected, redirects to speech capture.

Parameter	Source	Description
q	Query string	Question index
Digits	Twilio	Pressed DTMF digit (1-9)
CallSid	Twilio	Call identifier

Response: TwiML — either redirects to next question or to `/mcqo-other-handler`

POST `/mcqo-other-handler?q={index}`

MCQO "Other" Speech Handler

Captures (but does not store) the free-speech "Other" response and moves to the next question.

Parameter	Source	Description
q	Query string	Question index
SpeechResult	Twilio	Free-speech "Other" response

Response: TwiML — redirects to `/next?q={q+1}`

POST `/call-status`

Twilio Call Status Webhook

Receives call completion events. Updates participant state and triggers Excel export on completion.

Parameter	Source	Description
CallSid	Twilio	Call identifier
CallStatus	Twilio	One of: <code>completed</code> , <code>no-answer</code> , <code>busy</code> , <code>failed</code> , <code>canceled</code>

Response: `200 OK`

POST `/recording-done`

Twilio Recording Completion Webhook

Downloads the completed recording WAV and queues it for ML processing.

Parameter	Source	Description
CallSid	Twilio	Call identifier
RecordingUrl	Twilio	URL to download recording
RecordingStatus	Twilio	<code>completed</code> or other

Response: `200 OK`

GET `/health`

Health Check

Simple liveness probe.

Response: 200 OK — body: ok

POST /conference_host, POST /conference_join, POST /conference_ivr, POST /conference_ivr_next

Conference Call Endpoints

Handle the Twilio conference call flow. The host enters a waiting room where IVR questions play, then the conference starts when all questions are complete.

POST /silence

Silence Generator

Returns TwiML with a 60-second pause. Used as conference hold music fallback.

GET /ivr-audio/<filename>

Static IVR Audio Server

Serves pre-generated Azure TTS audio files from data/ivr_audio/.

5.2 Authenticated Endpoints (Login Required)

All /admin/* routes require an active session (cookie-based) or a valid ADMIN_TOKEN query parameter.

GET /admin

Admin Dashboard

Renders the full admin dashboard HTML page.

Response: 200 OK — HTML page

GET /admin/live_state

Live State JSON API

Returns current participant state for live-polling dashboard updates.

Response:

```
{
  "total": 42,
  "counts": {
    "pending": 10,
    "in_progress": 5,
    "completed": 25,
    "failed": 2
  },
  "participants": [
    {
      "participant_id": "P001",
      "phone_masked": "+2*****1234",
      "status": "completed",
      "attempts": 1,
      "engaged": true,
```

```
    "scheduled_local": "2026-03-15 14:00",
    "scheduled_input": "2026-03-15 14:00"
  }
]
}
```

POST /admin/upload_contacts

Upload Participant CSV

Accepts a CSV file with headers `participant_id,phone_e164`. Creates participant entries with `idle` status (not immediately callable).

POST /admin/save_questions

Save Survey Questions

Saves the survey question text to `data/questions.txt`.

POST /admin/schedule

Schedule Participant Call

Sets a participant's scheduled call time (NYC timezone).

Parameter	Source	Description
<code>participant_id</code>	Form	Participant ID
<code>local_time</code>	Form	<code>YYYY-MM-DD HH:MM</code> (NYC time)

POST /admin/dial_now

Force Dial

Triggers `run_once(force=True)` — calls all eligible participants immediately, bypassing schedule/retry timing.

POST /admin/pause / POST /admin/resume

System Pause / Resume

Toggles the global scheduler pause state.

POST /admin/reset_state

Reset State

Creates timestamped backups of `participants.json` and `call_log.csv`, then resets all state.

GET /admin/export_excel

Export Survey Responses (Original Language)

Downloads `ivr_responses.xlsx` containing MCQ/MCQO responses with option text decoded.

GET /admin/export_excel_english

Export Survey Responses (English)

Translates all response cells to English and downloads `ivr_responses_english.xlsx`.

POST /admin/conference_call

Start Conference Call

Dials two phone numbers into a shared Twilio conference room.

Parameter	Source	Description
<code>number_1</code>	Form	First phone number (E.164)
<code>number_2</code>	Form	Second phone number (E.164)

5.3 Authentication Endpoints

GET /login

Login Page

Renders the login form.

POST /login

Login Submit

Authenticates the user against `config.yaml` password hashes.

Parameter	Source	Description
<code>username</code>	Form	Username (case-insensitive)
<code>password</code>	Form	Plaintext password

Responses:

- Success: redirect to `/admin`
- Invalid credentials: re-render login page with error
- Locked out: show remaining lockout time

POST /logout

Logout

Clears the session and logs the event.

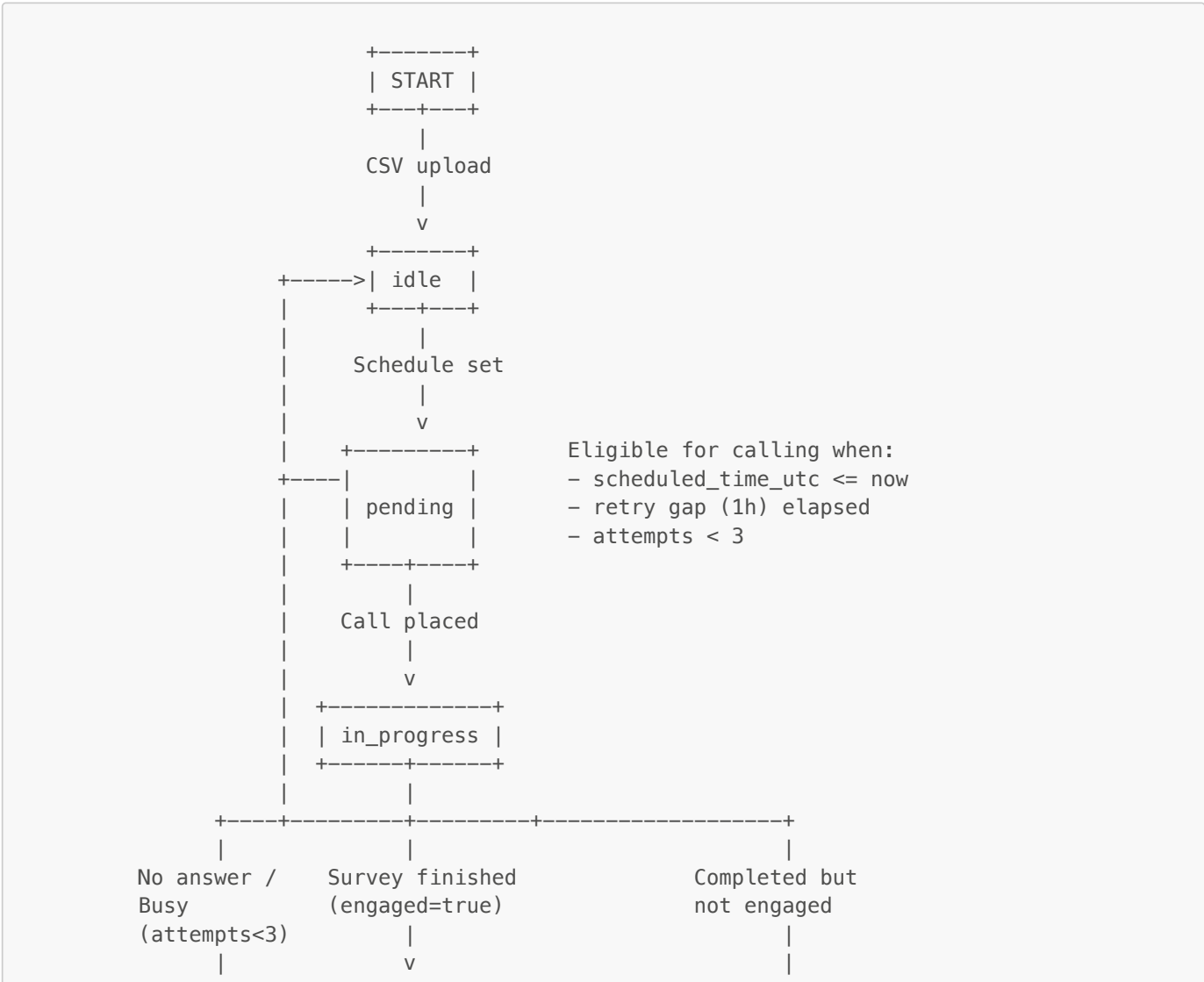
6. Data Models & State Schema

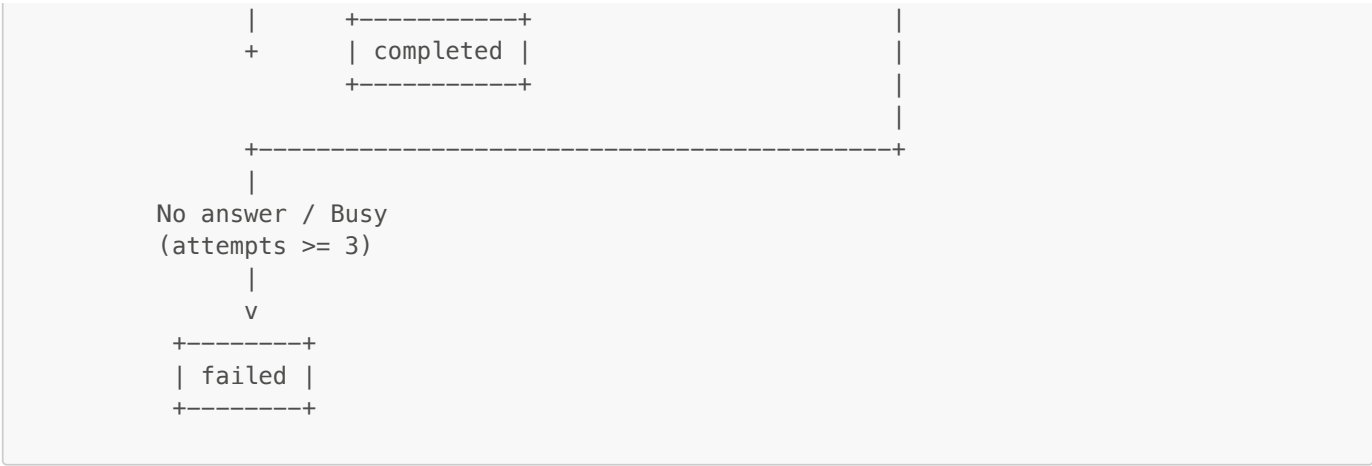
6.1 Participant Schema

Each participant in `data/state/participants.json` follows this schema:

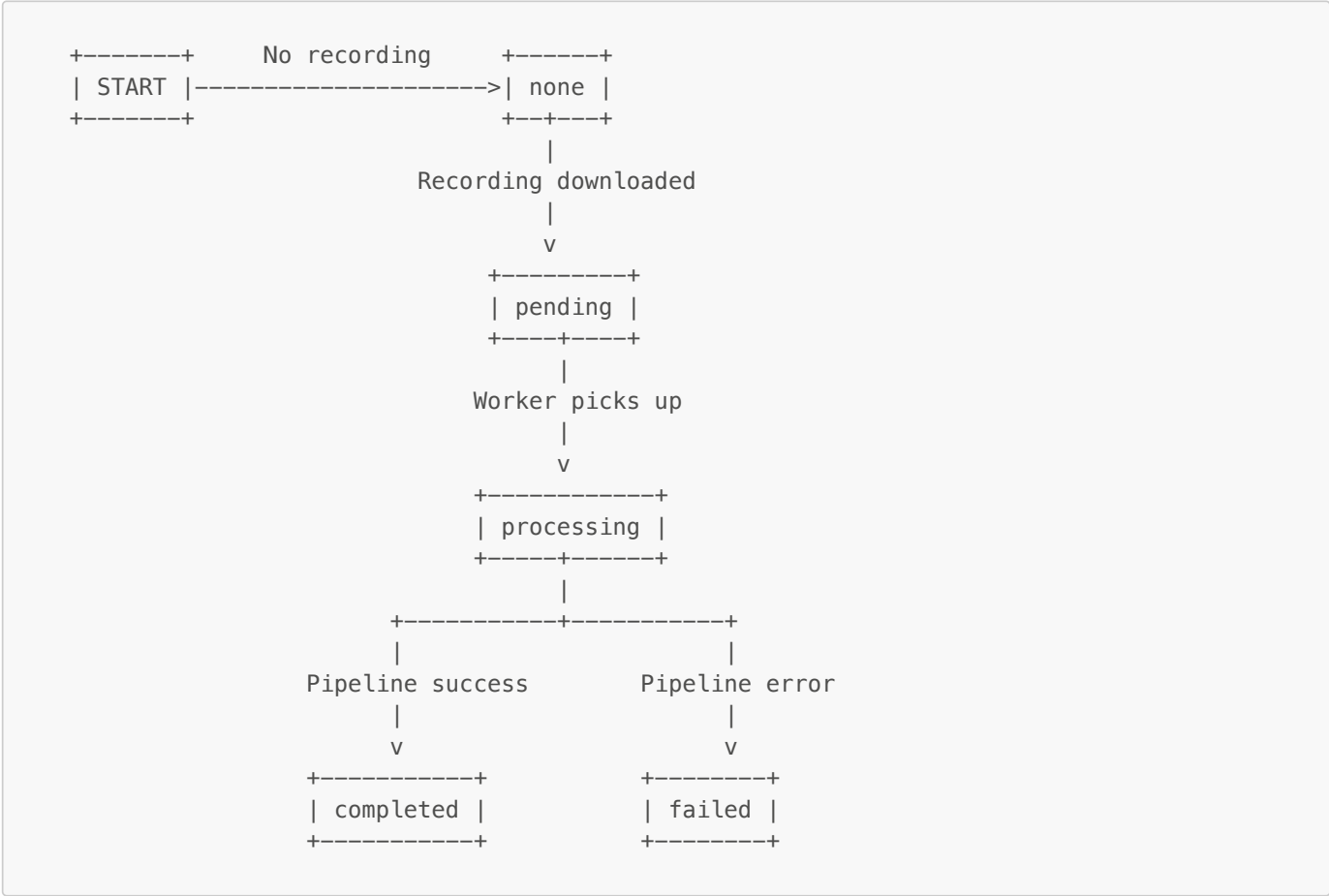
```
{
  "P001": {
    "status": "pending",
    "attempts": 0,
    "last_call_time": null,
    "last_call_sid": null,
    "last_call_status": null,
    "engaged": false,
    "last_recording_url": null,
    "last_outputs": {},
    "scheduled_time_local": null,
    "scheduled_time_utc": null,
    "phone_e164": "+254700000000",
    "responses": {
      "q1": "2",
      "q2": "1",
      "q3": "3"
    },
    "survey_q_counter": 3,
    "processing_status": "completed",
    "audio_path": "data/audio/P001_20260315_140000.wav",
    "recording_url": "https://api.twilio.com/2010-04-01/Accounts/.../Recordings/RE..."
  }
}
```

6.2 Participant State Machine





6.3 Processing Status State Machine



6.4 Call Log Schema

data/state/call_log.csv columns:

Column	Type	Description
timestamp_utc	ISO 8601	Event timestamp
participant_id	String	Participant identifier
phone_masked	String	Masked phone (e.g., +2*****1234)
direction	String	inbound or outbound-api
call_sid	String	Twilio call SID
recording_url	URL	Twilio recording URL

Column	Type	Description
audio_path	String	Local file path
transcript_path	String	Local transcript path
translation_path	String	Local translation path
english_audio_path	String	Local English audio path

6.5 Auth State Schema

data/auth_state.json:

```
{
  "fails": {
    "username|ip_address": [1711234567, 1711234580]
  },
  "locks": {
    "username|ip_address": 1711235467
  }
}
```

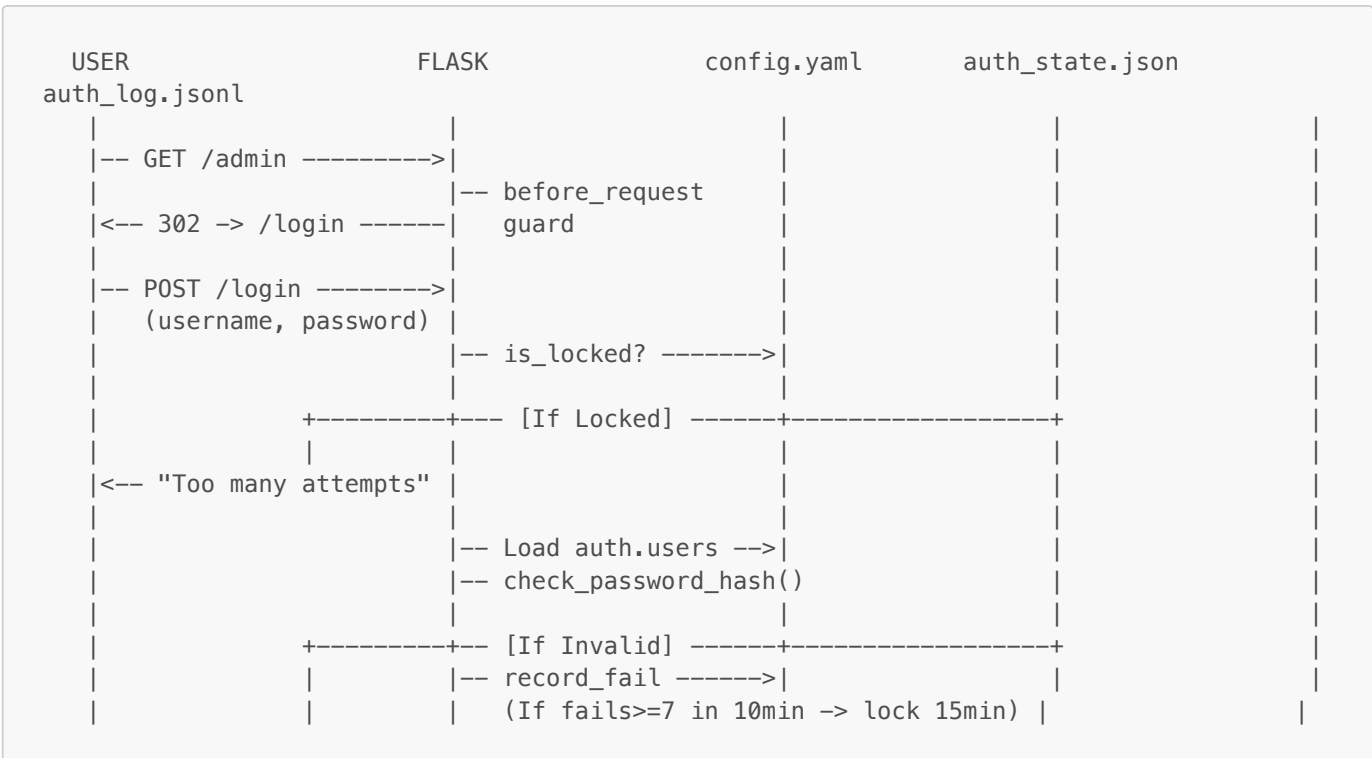
6.6 Settings Schema

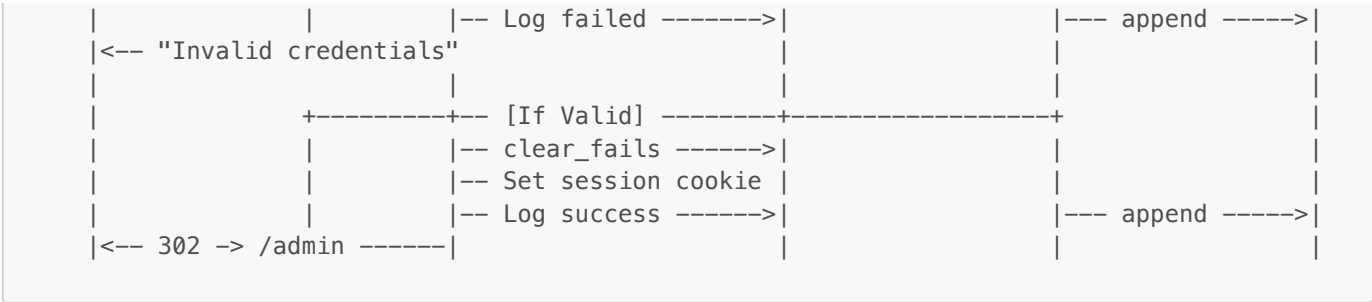
data/state/settings.json:

```
{
  "paused": false
}
```

7. Authentication & Security

7.1 Authentication Flow

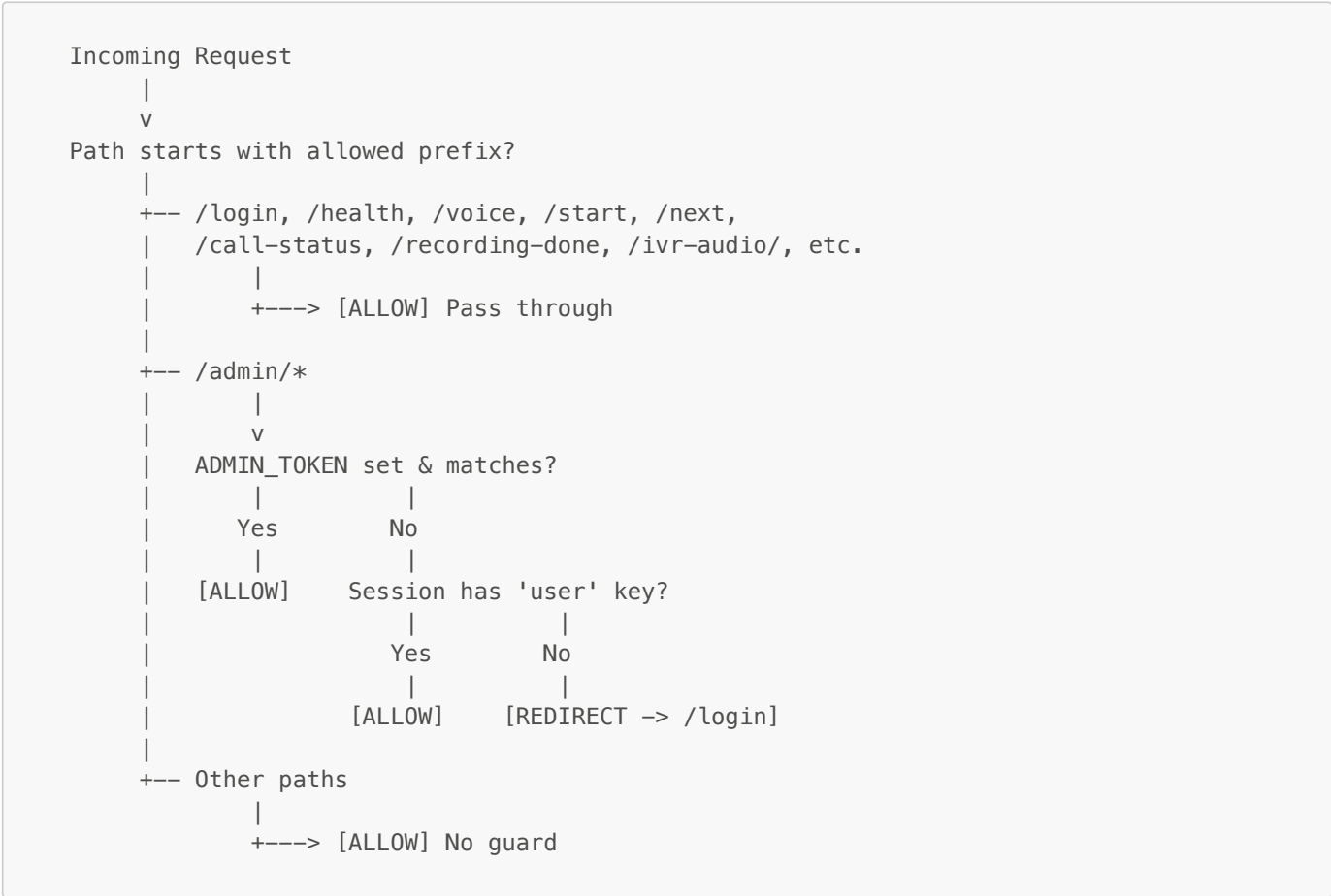




7.2 Security Measures

Measure	Implementation	Details
Password Hashing	PBKDF2-SHA256	1,000,000 iterations via Werkzeug
Brute-Force Protection	Rate limiting	7 failures in 10 minutes -> 15-minute lockout per username+IP
Session Security	Flask sessions	HttpOnly, SameSite=Lax, Secure=True, 8-hour lifetime
Phone Masking	mask_phone()	Only first 2 + last 4 digits shown (e.g., +2*****1234)
PII Protection	.gitignore	Contacts CSV, participant state, audio, and logs are git-ignored
Route Protection	@app.before_request	All /admin routes require session or admin token
Webhook Passthrough	Allowlist	Twilio webhook paths (/voice, /next, etc.) bypass auth
Audit Logging	JSONL	Every login/logout event logged with IP, user-agent, timestamps

7.3 Request Guard Logic



8. Survey Question Engine

8.1 Question File Format

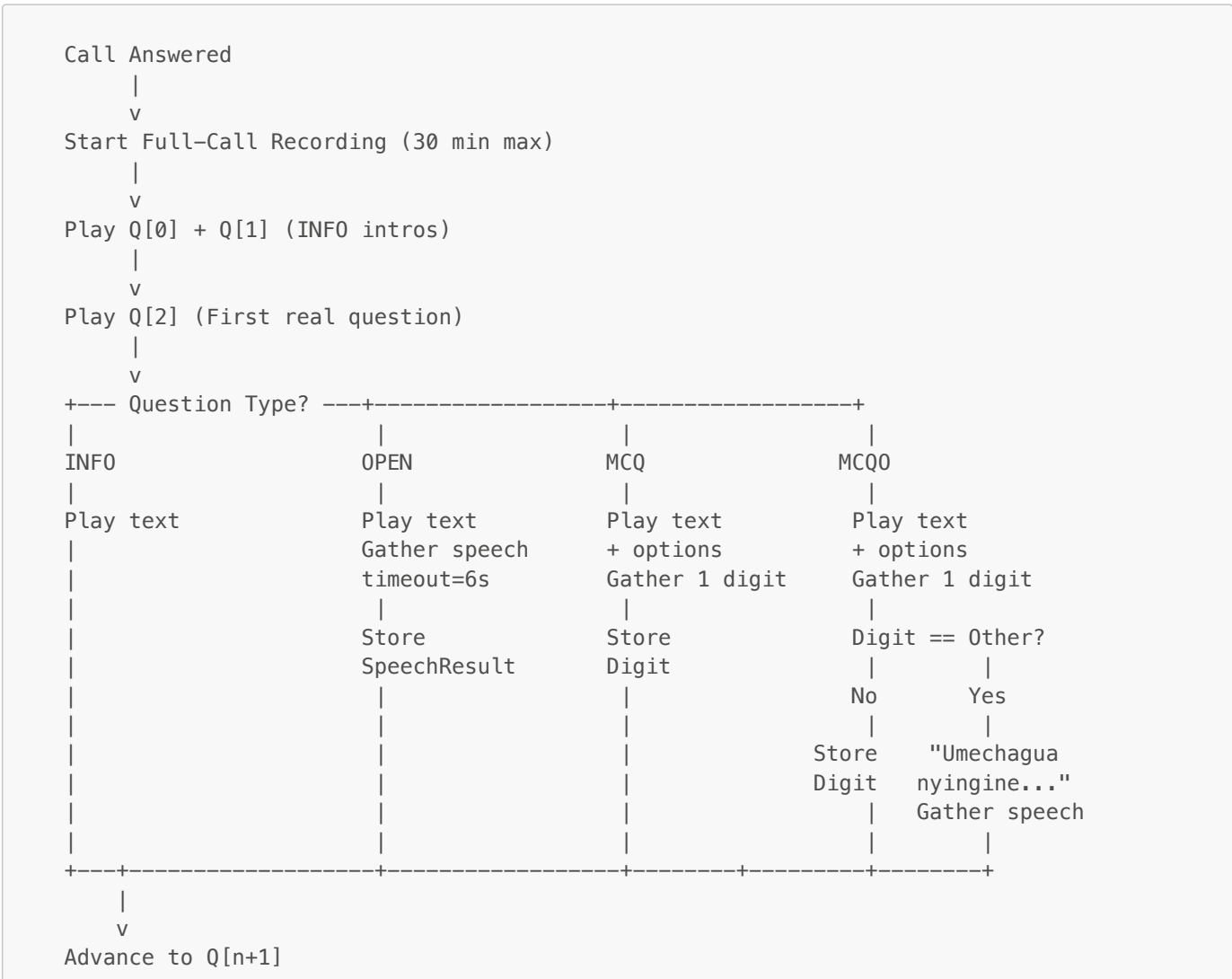
Questions are stored in `data/questions.txt` using a pipe-delimited format:

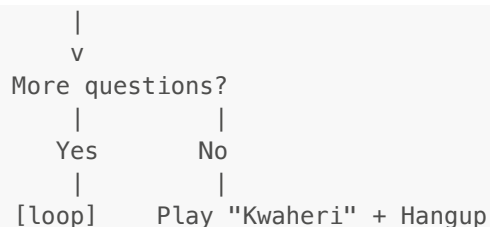
```
TYPE|Question Text|Option1|Option2|...
```

8.2 Supported Question Types

Type	Format	Input Method	Response Storage	Example
INFO	INFO\ text	None (read-only)	Not stored	INFO\ Maswali sehemu ya kwanza
OPEN	OPEN\ text	Speech (<Gather input="speech">)	Raw speech text	OPEN\ Tafadhali sema jina lako
MCQ	MCQ\ text\ opt1\ opt2\ opt3	DTMF keypad (<Gather input="dtmf">)	Digit (1-9)	MCQ\ Nani..? \ Marafiki\ Mume\ Watoto
MCQO	MCQO\ text\ opt1\ opt2\ Nyingine	DTMF + optional speech	Digit (1-9)	MCQO\ Kupanga..? \ Ndiyo\ Hapana\ Nyingine

8.3 IVR Question Flow





8.4 TTS Prompt Generation

Survey prompts are spoken aloud using **Azure Cognitive Services Neural TTS**:

- **Voice:** `sw-KE-ZuriNeural` (Swahili) / `en-US-JennyNeural` (English)
- **Rate:** `-15%` prosody for clearer, slower speech
- **Caching:** SHA1 hash of `voice|format|text` to disk-cached MP3 in `data/ivr_audio/`
- **Serving:** Public URL via `/ivr-audio/{hash}.mp3`, referenced in TwiML `<Play>` tags
- **Format:** SSML with XML entity escaping

8.5 MCQ Option Verbalization

For MCQ/MCQO questions, options are read aloud with their digit mapping:

```
"finya 1 kwa Ndiyo. finya 2 kwa Hapana. finya 3 kwa Nyingine."
```

(Translation: "press 1 for Yes. press 2 for No. press 3 for Other.")

9. Configuration Reference

9.1 Environment Variables (`.env`)

Variable	Required	Default	Description
<code>TWILIO_ACCOUNT_SID</code>	Yes	--	Twilio account SID
<code>TWILIO_AUTH_TOKEN</code>	Yes	--	Twilio auth token
<code>TWILIO_FROM_NUMBER</code>	Yes	--	Twilio phone number (E.164)
<code>PUBLIC_BASE_URL</code>	Yes	--	Public URL for webhooks (ngrok or custom)
<code>ADMIN_TOKEN</code>	No	""	Optional token for admin access without login
<code>AZURE_SPEECH_KEY</code>	Yes	--	Azure Cognitive Services subscription key
<code>AZURE_SPEECH_REGION</code>	Yes	--	Azure region (e.g., <code>eastus</code>)
<code>AZURE_TTS_VOICE_SW</code>	No	<code>sw-KE-ZuriNeural</code>	Swahili TTS voice name
<code>AZURE_TTS_VOICE_EN</code>	No	<code>en-US-JennyNeural</code>	English TTS voice name
<code>AZURE_TTS_FORMAT</code>	No	<code>audio-16khz-128kbitrate-mono-mp3</code>	Audio output format
<code>MAX_CALLS_PER_TICK</code>	No	3	Max concurrent calls per scheduler tick
<code>CALL_SPACING_SEC</code>	No	0.8	Delay between outbound calls

Variable	Required	Default	Description
FLASK_SECRET_KEY	No	CHANGE_ME_NOW	Flask session signing key
AUTO_START_NGROK	No	1	Auto-start ngrok tunnel
APP_OPEN_URL	No	--	Custom URL to open in browser
OPEN_PUBLIC_URL	No	0	Open ngrok URL instead of localhost
AUTH_STATE_PATH	No	data/auth_state.json	Auth state file location
AUTH_LOG_PATH	No	data/auth_log.jsonl	Auth event log location
AUTH_MAX_FAILS	No	7	Failed attempts before lockout
AUTH_LOCK_SECONDS	No	900	Lockout duration (seconds)
AUTH_WINDOW_SECONDS	No	600	Failure tracking window (seconds)

9.2 Configuration File (config.yaml)

```
# Twilio Configuration
twilio:
  account_sid: ""           # (Overridden by .env)
  auth_token: ""           # (Overridden by .env)
  from_number: ""          # (Overridden by .env)
  public_base_url: ""      # (Overridden by .env)

# IVR Configuration
ivr:
  questions_file: "data/questions.txt"  # Path to questions file
  gather_timeout_sec: 6                 # Seconds to wait for input
  speech_timeout: "auto"               # Twilio auto speech detection
  say_voice: "alice"                   # Fallback Twilio voice
  speech_language: "sw-KE"             # Kiswahili-Kenya for recognition

# Audio Preprocessing (DeepFilterNet)
audio_processing:
  enabled: true                       # Toggle noise removal
  backend: "deepfilternet"            # Only supported backend
  processed_dir: "data/audio_processed" # Cleaned audio output
  temp_dir: "data/audio_processed/tmp" # Intermediate files
  output_sample_rate: 16000           # Whisper expects 16kHz
  model_sample_rate: 48000            # DeepFilterNet expects 48kHz
  channel_mode: "mixdown"             # "mixdown" or "channel"
  caller_channel: 0                   # Channel index if mode="channel"
  keep_intermediate_files: false      # Cleanup temp files

# Authentication
auth:
  users:
    username:
      password_hash: "pbkdf2:sha256:..." # Werkzeug generate_password_hash()
```

9.3 State Management Constants

Constant	Value	Location	Description
MAX_ATTEMPTS	3	state.py	Maximum call attempts per participant
RETRY_GAP	1 hour	state.py	Minimum time between retry attempts

Constant	Value	Location	Description
RECORDING_MAX_SEC	1800 (30 min)	twilio_handler.py	Maximum recording duration
GATHER_TIMEOUT	6 seconds	config.yaml	DTMF/speech input timeout
SCHEDULER_INTERVAL	15 seconds	twilio_handler.py	Scheduler polling interval
WORKER_POLL	5 seconds	background_worker.py	Worker polling interval
MAX_CHARS	3000	translate.py	Max characters per translation chunk

10. Admin Dashboard

10.1 UI Layout

AudioSurvey AI -- AdminSystem: [RUNNING]
NYC time: 2026-03-20 14:32:15 EDTLogged in as: krishnanand
[Sign out]

[Dial Now] [Start] [Stop] [State Refresh] [Export Excel] [English]

42
Total

10
Pending

5
In Progress

25
Completed

Upload Contacts
[Choose CSV][Upload]

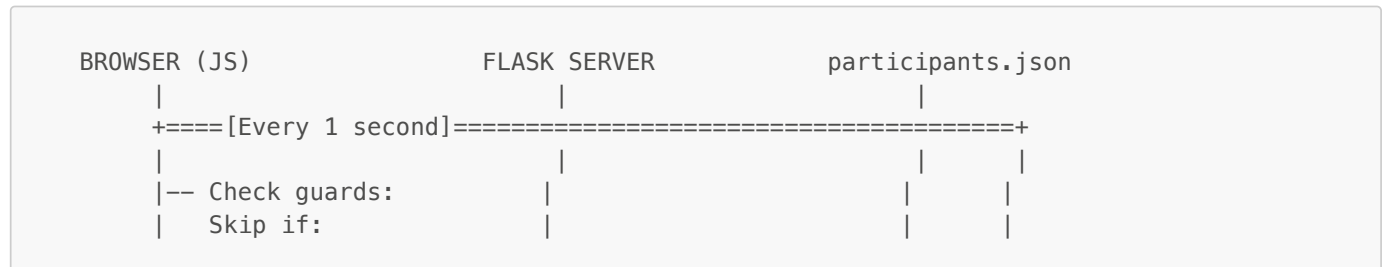
Questions
+-----+
INFO|Maswali kuhu...
OPEN|Tafadhali se...
MCQ|Question text...
+-----+
[Save questions]

Conference Call
[+1...] [+1...] [Start call]

Participants

ID	Phone	Status	Att.	Engaged	Sched.	Schedule
P001	+2****34	Completed	1	Engaged	03-15	[_] [Set]
P002	+2****56	Pending	0	Not eng.		[_] [Set]
P003	+2****78	Failed	3	Not eng.	03-14	[_] [Set]

10.2 Live Polling Architecture



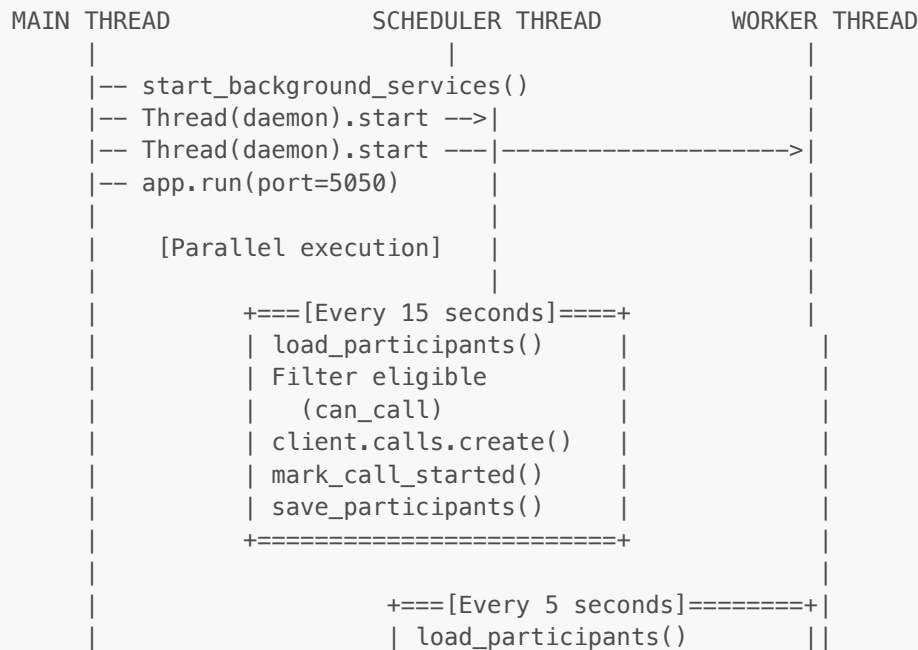
```
| - Flatpickr open | | | | |
| - Schedule input dirty | | | | |
| - Active focus in table | | | | |
| - Previous poll active | | | | |
|
|-- GET /admin/live_state --> | | | | | |
|                               |-- load_participants --> | | | | |
|                               <-- JSON state -----> | | | | |
| <-- {total, counts, | | | | |
|     participants} | | | | |
|
|-- Update KPI cards | | | | |
|-- Re-render table rows | | | | |
|-- Re-initialize Flatpickr | | | | |
|
+=====+
```

10.3 Dashboard Features

Feature	Description
Live NYC Clock	JavaScript Intl.DateTimeFormat, updates every second
KPI Cards	Total, Pending, In Progress, Completed — live-updated
Status Pills	Color-coded badges: green=completed, yellow=in_progress, red=failed, gray=pending
Engaged Badges	Green dot = engaged, gray dot = not engaged
Flatpickr Date Picker	Dark-themed calendar with time selection and confirm plugin
Dirty Input Detection	Prevents live-poll from overwriting user's in-progress schedule edits
File Upload	Styled file input with chosen filename display
Dark Theme	Custom CSS with glassmorphism, radial gradients, and blur effects

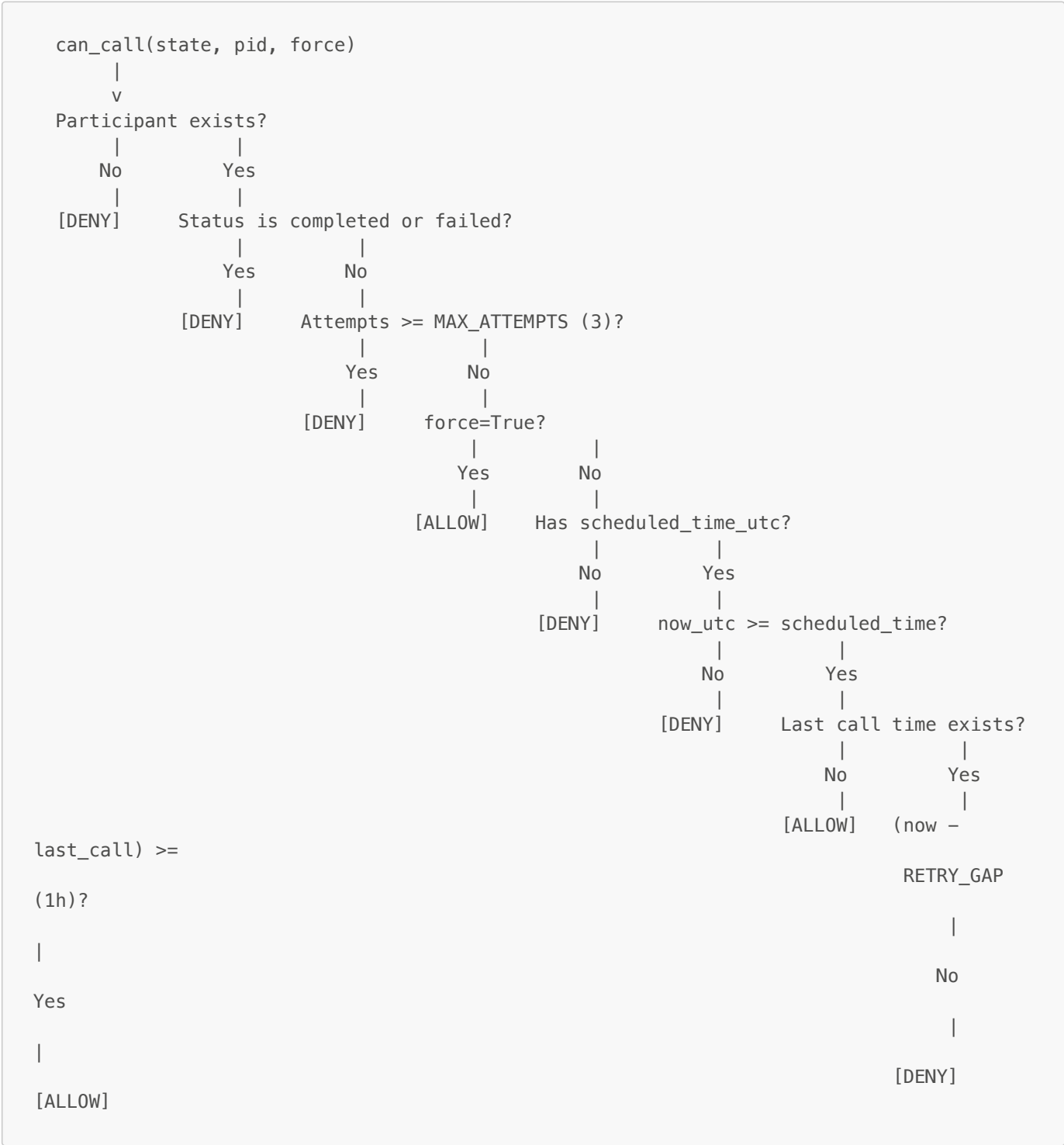
11. Background Services

11.1 Service Lifecycle



```
|
|
| Find processing_status ||
| = "pending" ||
| Run ML pipeline ||
| mark_completed() ||
| save_participants() ||
|=====+
```

11.2 Call Eligibility Logic (can_call)



11.3 Worker Pipeline Stages

Stage	Progress	Duration	Description
1. Prepare	10-25%	~5s	FFmpeg: extract channel, resample to 48kHz mono

Stage	Progress	Duration	Description
2. Denoise	25-60%	~30-60s	DeepFilterNet: neural noise removal (GPU if available)
3. Resample	60-75%	~3s	FFmpeg: downsample to 16kHz for Whisper
4. Transcribe	75-85%	~60-120s	Whisper large-v3: Swahili STT
5. Translate	85-93%	~5-15s	Google Translate: chunked sw to en
6. English TTS	93-97%	~5-10s	gTTS: English audio synthesis
7. Complete	100%	--	Mark completed, save state

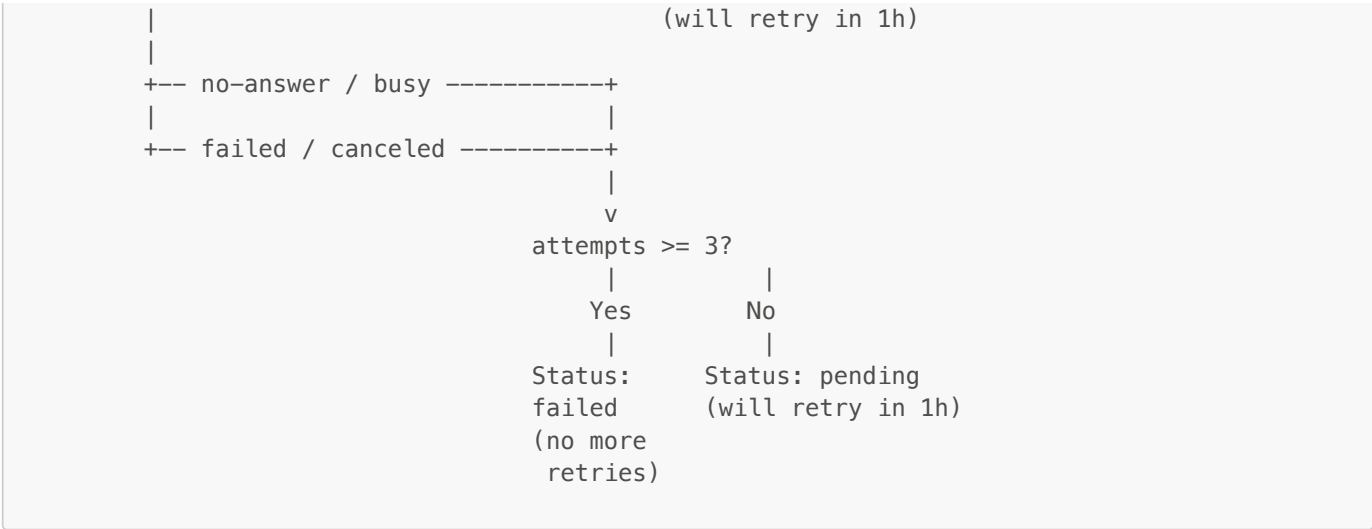
12. Error Handling & Resilience

12.1 Fault Tolerance Matrix

Component	Failure Mode	Handling Strategy
Twilio Call	No answer / Busy	Mark pending , retry after 1h (up to 3 attempts)
Twilio Call	Failed / Canceled	Mark failed if max attempts reached
Azure TTS	API error	RuntimeError raised, prevents corrupted audio serving
Azure TTS	Duplicate request	Disk cache by content hash prevents re-synthesis
Whisper	Transcription drift	condition_on_previous_text=False, temperature=0.0
Google Translate	API error	3 retries with 1.5s backoff; failed chunks marked with [TRANSLATION_FAILED_CHUNK]
Google Translate	Returns None	Explicit None check with RuntimeError
DeepFilterNet	Processing error	Graceful fallback to unprocessed audio
Recording Download	HTTP error	Logged, returns 200 (Twilio won't retry)
State File	Corrupt JSON	Renamed to .corrupt , returns empty state
State File	Concurrent access	threading.RLock() + atomic write (.tmp then os.replace)
Scheduler	Exception in tick	Caught and logged, loop continues
Worker	Exception in pipeline	Caught, processing_status set to failed , loop continues
Auth State	Missing file	Returns safe defaults {"fails": {}, "locks": {}}
Config	Missing config.yaml	Uses hardcoded defaults

12.2 Retry Policy

```
Call Placed (attempt #N)
|
v
Call Result
|
+-- completed + engaged -----> Status: completed
|                               Pipeline: triggered
|
+-- completed + NOT engaged -----> Status: pending
```



13. Logging & Observability

13.1 Logging Architecture

Log Type	Format	Location	Purpose
Application Log	Colored console	stdout	Real-time operational visibility
Auth Event Log	JSONL	data/auth_log.jsonl	Security audit trail
Call Log	CSV	data/state/call_log.csv	Call history and recording tracking
Worker Progress	Terminal progress bar	stdout	ML pipeline progress monitoring

13.2 Console Log Format

```
[NYC 2026-03-20T14:32:15-04:00 | UTC 2026-03-20T18:32:15Z] PROMPT SENT | CallSid=CA123 | Participant=P001 | q3_mcq | Text="..."
```

13.3 Auth Event Log Format

```
{"event":"login","user":"krishnanand","ip":"127.0.0.1","login_utc":"2026-03-20T18:32:15Z","login_local":"2026-03-20 14:32:15 EDT","user_agent":"Mozilla/5.0..."}
{"event":"logout","user":"krishnanand","ip":"127.0.0.1","logout_utc":"2026-03-20T22:15:00Z","logout_local":"2026-03-20 18:15:00 EDT","session_duration_sec":13365}
```

13.4 Worker Progress Bar

```
[BackgroundWorker] participant=P001 | [#####-----] 80% | Removing background noise
```

13.5 Silenced Libraries

The following library loggers are suppressed to reduce noise:

httpx, httpcore, httpcore.connection, httpcore.http2, hpack, hpack.hpack, h2, h2.connection, h2.config, gtts, gtts.tts, googletrans, urllib3, torio, torio._extension, torio._extension.utils, torchaudio

14. Deployment Guide

14.1 Prerequisites

Requirement	Version	Purpose
Python	3.10+	Runtime
pip	--	Package management
ngrok	3.x	HTTPS tunneling
FFmpeg	4.x+	Audio processing
Twilio Account	--	Voice API
Azure Account	--	Cognitive Services TTS

14.2 Local Development Setup

```
# 1. Clone the repository
git clone https://github.com/krishnanand20/audiosurvey_ai.git
cd audiosurvey_ai

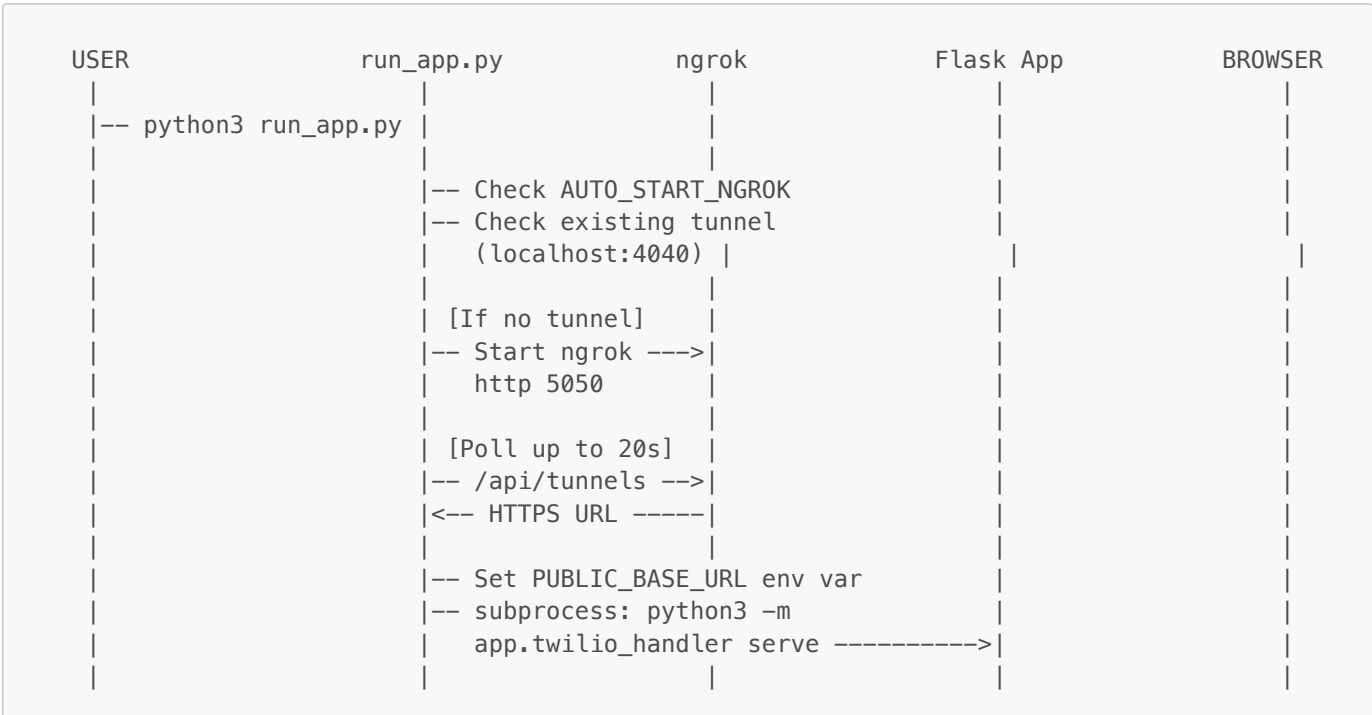
# 2. Create virtual environment
python3 -m venv venv
source venv/bin/activate

# 3. Install dependencies
pip install -r requirements.txt

# 4. Configure environment
cp .env.example .env
# Edit .env with your Twilio + Azure credentials

# 5. Launch (auto-starts ngrok + Flask + opens browser)
python3 run_app.py
```

14.3 Startup Sequence



```
| Load .env, config.yaml
| Load Whisper model (large-v3)
| start_background_services()
| app.run(port=5050)
|
|-- Wait 1.5s
|-- Open browser ---|----->
| http://127.0.0.1:5050/admin
```

14.4 macOS DMG Distribution

The application can be packaged as a native macOS **.app** inside a DMG installer:

```
cd packaging/macos_dmg
./build_macos_dmg.sh
# Output: output/AudioSurvey-AI.dmg
```

Build process:

1. Compiles Swift icon generator, generates 1024x1024 PNG icon
2. Creates `.iconset` with all required sizes via `sips`
3. Converts to `.icns` via `iconutil`
4. Builds `.app` bundle with `Info.plist`, launcher script, and project files
5. Creates read-write DMG, sets volume icon, converts to compressed DMG
6. Optionally embeds icon resource via `Rez`

App bundle structure:

```
AudioSurvey AI.app/  
  Contents/  
    Info.plist  
    MacOS/  
      AudioSurveyAI (bash launcher)  
    Resources/  
      AppIcon.icns  
      project/ (full project copy)
```

15. Directory Structure

```
audiosurvey_ai/
|
|-- .env                # Environment variables (secrets -- gitignored)
|-- .gitignore          # Git exclusion rules
|-- config.yaml         # Application configuration
|-- main.py             # Batch processing entry point
|-- run_app.py          # Application launcher (ngrok + Flask)
|-- requirements.txt     # Python dependencies
|-- README.md           # Quick-start documentation
|-- DOCUMENTATION.md    # This file
|
|-- app/                # Application source code
|   |-- twilio_handler.py # Flask app, IVR routes, Azure TTS, auth
|   |-- dashboard.py      # Admin dashboard UI + routes
|   |-- state.py          # Thread-safe state management
```

```
| |-- scheduler.py           # Background call scheduler
| |-- background_worker.py  # ML processing pipeline worker
| |-- audio_preprocess.py   # DeepFilterNet noise removal
| |-- transcribe.py         # Whisper speech-to-text
| |-- translate.py          # Google Translate integration
| |-- tts.py                # Google TTS (English audio)
| |-- azure_tts.py          # Azure Cognitive Services TTS
| |-- export_excel.py       # Excel response export
| |-- auth.py               # Authentication helpers
| |-- utils.py              # Scheduling utilities
| |-- file_naming.py        # Safe filename generation
| |-- logger.py             # Colored logging setup
| |-- runtime_warnings.py   # Warning suppression
| |-- twilio_utils.py       # Twilio call helpers
|
|-- data/                   # Runtime data (mostly gitignored)
| |-- questions.txt         # Survey questions (pipe-delimited)
| |-- state/                # Participant state + logs
| | |-- participants.json   # Participant records
| | |-- call_log.csv        # Call history
| | |-- settings.json       # System settings
| |-- audio/                # Raw call recordings (.wav)
| |-- audio_processed/      # Denoised recordings (.wav)
| |-- transcripts/          # Whisper transcriptions (.txt)
| |-- translations/         # English translations (.txt)
| |-- english_audio/        # English TTS audio (.mp3)
| |-- ivr_audio/            # Cached IVR prompts (.mp3)
| |-- results/              # Excel exports (.xlsx)
| |-- *.csv                 # Contact lists
|
|-- packaging/              # Distribution packaging
| |-- macos_icon_generator.swift # Programmatic icon generation
| |-- macos_dmg/
| | |-- build_macos_dmg.sh   # DMG build script
| | |-- README.md            # Build instructions
| | |-- output/
| | | |-- AudioSurvey-AI.dmg # Built DMG installer
```

16. Dependency Matrix

16.1 Core Dependencies

Package	Version	Purpose	Critical?
Flask	3.1.2	Web framework	Yes
twilio	9.10.0	Telephony API	Yes
openai-whisper	20240930	Speech-to-text	Yes
azure-cognitiveservices-speech	1.48.1	Neural TTS (prompts)	Yes
googletrans	4.0.0rc1	Translation (sw to en)	Yes
gTTS	2.5.4	English audio generation	Yes
deepfilternet	0.5.6	Background noise removal	Medium
torch	2.8.0	ML framework (Whisper + DeepFilterNet)	Yes
torchaudio	2.8.0	Audio tensor operations	Yes

16.2 Data & Export

Package	Version	Purpose
pandas	2.3.3	DataFrame operations for Excel export
openpyxl	3.1.5	Excel file writing
numpy	1.26.4	Numerical operations
pydub	0.25.1	Audio manipulation

16.3 Infrastructure

Package	Version	Purpose
python-dotenv	1.2.1	.env file loading
PyYAML	6.0.3	config.yaml parsing
unicorn	23.0.0	Production WSGI server
Werkzeug	3.1.4	Password hashing, HTTP utilities
colorlog	6.10.1	Colored console logging
requests	2.32.5	HTTP client (recording download)
Jinja2	3.1.6	Template rendering

16.4 External Services

Service	Provider	Purpose	Pricing Model
Voice API	Twilio	Outbound/inbound calls, recording, DTMF/speech gather	Per-minute
Speech TTS	Azure Cognitive Services	Neural TTS for IVR prompts (Swahili + English)	Per-character
Translation	Google Translate	Kiswahili to English translation	Free (scraping)
Text-to-Speech	Google TTS (gTTS)	English audio from translations	Free
Tunneling	ngrok	HTTPS tunnel for Twilio webhooks	Free tier available

17. Glossary

Term	Definition
IVR	Interactive Voice Response — automated phone menu system
DTMF	Dual-Tone Multi-Frequency — keypad tones (pressing 1-9, *, #)
Twiml	Twilio Markup Language — XML instructions for call handling
TTS	Text-to-Speech — converting text to spoken audio
STT	Speech-to-Text — converting spoken audio to text
E.164	International phone number format (e.g., +254700000000)
CallSid	Unique Twilio identifier for a phone call

Term	Definition
SSML	Speech Synthesis Markup Language — XML for controlling TTS output
DeepFilterNet	Neural network for real-time speech enhancement/noise removal
Whisper	OpenAI's multilingual speech recognition model
MCQ	Multiple Choice Question (DTMF input)
MCQO	Multiple Choice Question with "Other" (DTMF + optional speech)
OPEN	Open-ended question (speech input)
INFO	Informational prompt (no input collected)
Engaged	Participant spoke during the call (real speech detected)
NYC / NY_TZ	New York City timezone (America/New_York) used for scheduling
ngrok	Tunneling service exposing localhost to the internet
Kiswahili / sw	Swahili language (ISO 639-1: sw)
sw-KE	Swahili as spoken in Kenya (BCP 47 locale tag)
Pipeline	The 4-stage ML processing chain: denoise, transcribe, translate, TTS
Participant	A survey respondent identified by participant_id
Flatpickr	JavaScript date/time picker library used in the dashboard
PII	Personally Identifiable Information (phone numbers, names)
PBKDF2	Password-Based Key Derivation Function 2 (hashing algorithm)

End of Document

AudioSurvey AI v1.1.0 | Generated 2026-03-20